E-Commerce Marketplace: Commercial Performance Analysis

A business analytics case study using Excel, Power BI, and DAX

A commercial performance review designed around a key business question: **is revenue growth translating into profitable growth?** The analysis compares commercial performance across time, product categories, products, and acquisition channels.



2.46M	1.17M	47.65%	2,000
Revenue	Product contribution profit	Profit margin	Distinct customers

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1. Business Context & Analytical Approach

Revenue growth alone does not establish whether a business is becoming more profitable. This project provides a Chief Commercial Officer (CCO) view of revenue, product-level contribution, category economics, customer performance, and acquisition-channel trends.

Business questions

• Are revenue and profit moving together over time?

• How do revenue, quantity, profit, and margin differ across product categories?

• Which products combine high scale with comparatively lower margins?

• How do acquisition channels compare on revenue, profit, and order volume?

• Where is additional commercial investigation warranted?

Dataset overview

Table	Records
Customers	2,000
Orders	12,000
Order Items	30,120
Products	120
Events	59,599
Total (approx.)	**104,000**

Tools & techniques

Excel: data preparation and validation. **Power BI:** data modelling, report development, and visualization. **DAX:** KPI calculations and analytical measures. **Business analysis:** KPI interpretation, performance comparisons, and recommendations.

Data validation performed

The dataset was reviewed for missing values and duplicate IDs; relationships between tables were checked; order-status distribution was reviewed; date/time fields were converted to appropriate formats; and product cost was checked against catalogue price.

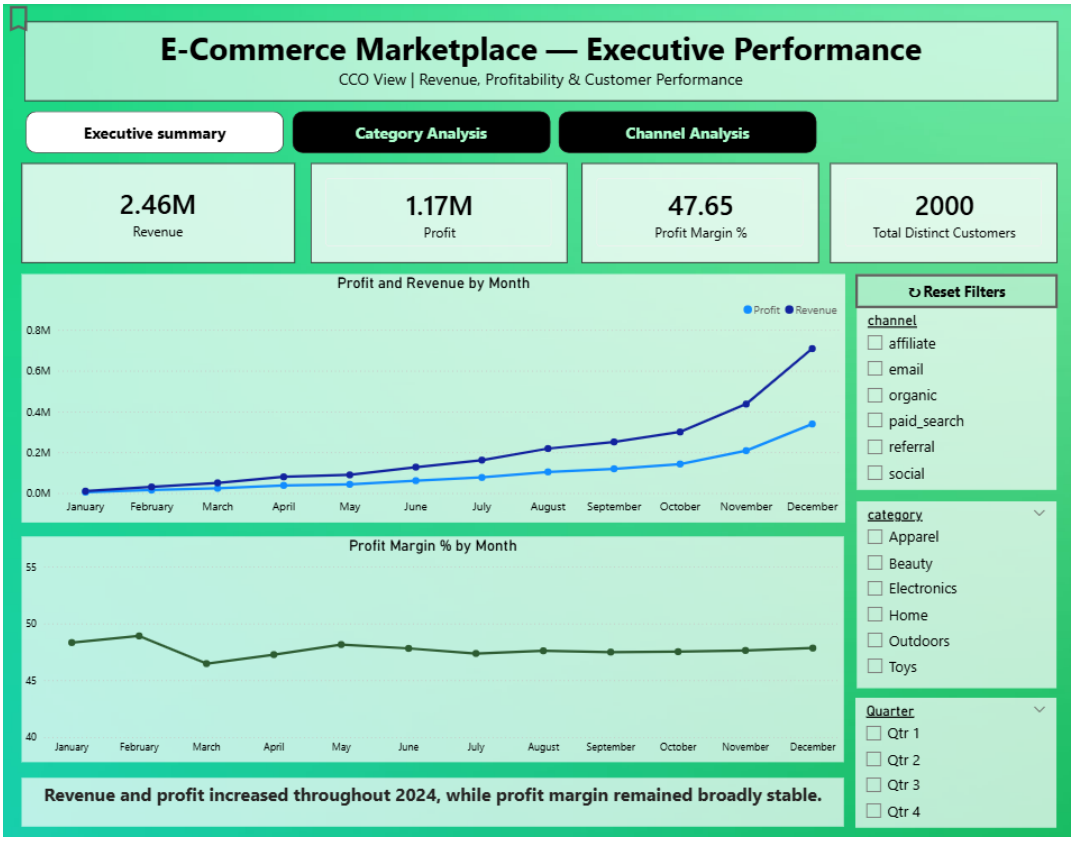
Order status distribution: Completed 66.8% · Cancelled 16.7% · Refunded 16.5%.

2. Measurement Rules & Executive Performance

Business rules

Revenue and profit calculations use **completed orders only**. Revenue is calculated as quantity multiplied by actual transaction price. Product contribution profit is calculated as (selling price – product unit cost) multiplied by quantity. Profit margin is profit divided by revenue, multiplied by 100.

The report includes filters for acquisition channel, product category, and quarter, supporting comparisons across the selected segments.

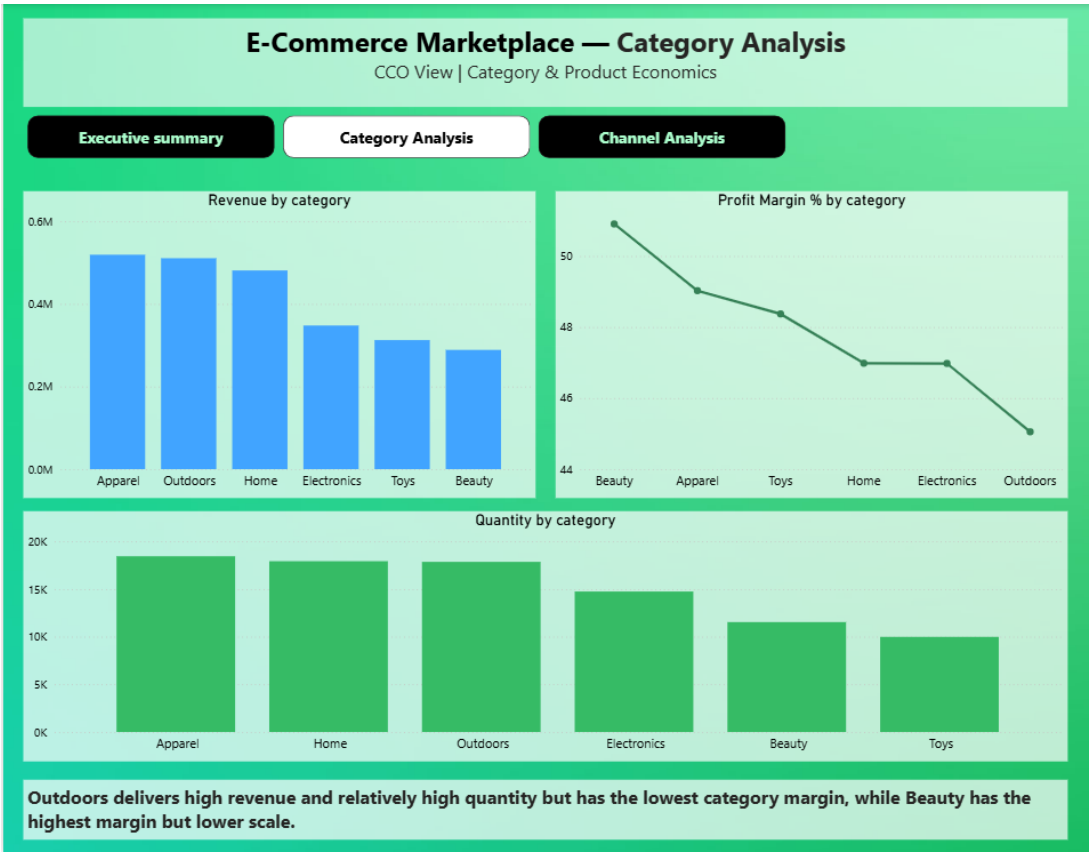


What the executive view indicates

The dashboard shows revenue and profit increasing through 2024, while monthly profit margin remains broadly stable. This does not indicate a sustained company-wide profitability decline in the period shown. The pattern makes category- and product-level comparisons important: overall stability can coexist with meaningful differences between parts of the business.

Interpretation note: the profit measure is product-level contribution based on product cost, not net profit.

3. Category Performance



Key finding

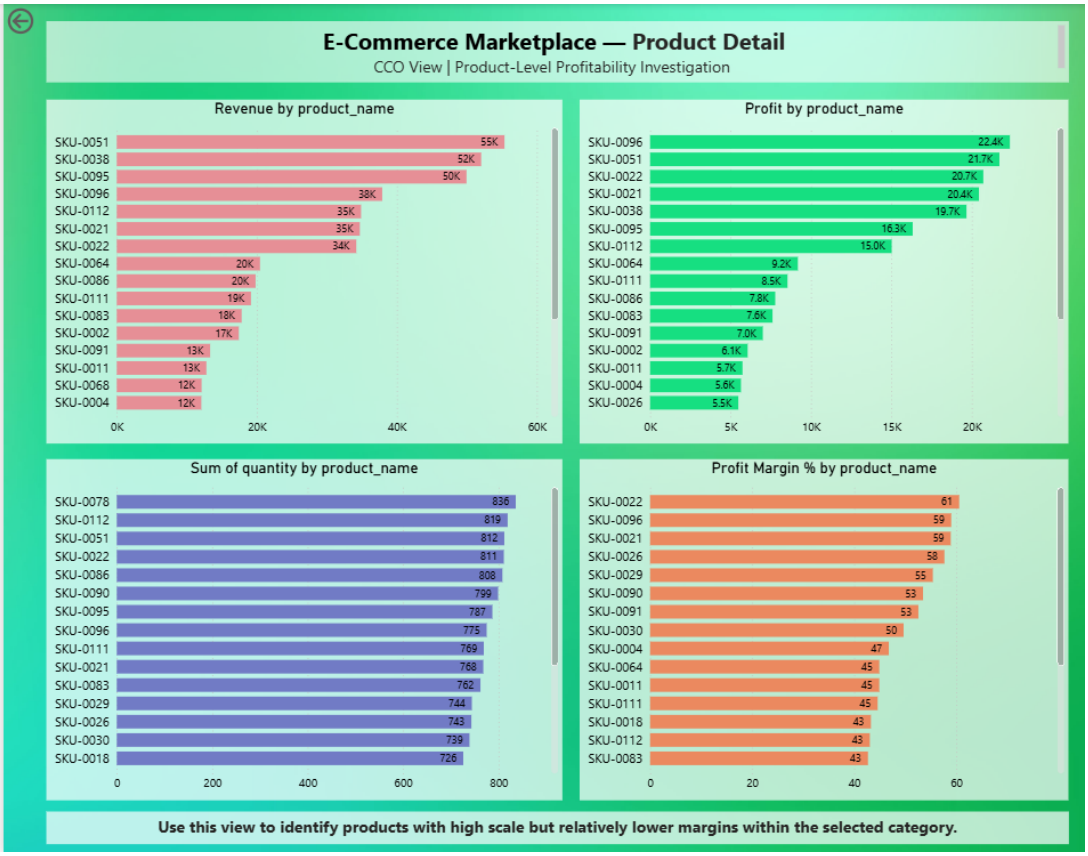
Outdoors generates high revenue and relatively high quantity, but has the lowest category-level profit margin in the report. **Beauty** has the highest margin, but lower overall scale. This illustrates why revenue rank alone is not enough to assess commercial performance.

Business interpretation

Outdoors is a reasonable priority for a targeted margin review because its scale makes the category commercially important and its margin is comparatively low. Beauty can provide a useful comparison point for understanding what differs in product mix, pricing, or unit economics, but the dashboard alone does not establish the cause of the margin gap.

Next question: which products contribute most to Outdoors revenue and profit, and do their margins suggest different review priorities?

4. Product-Level Investigation



Products identified for further review

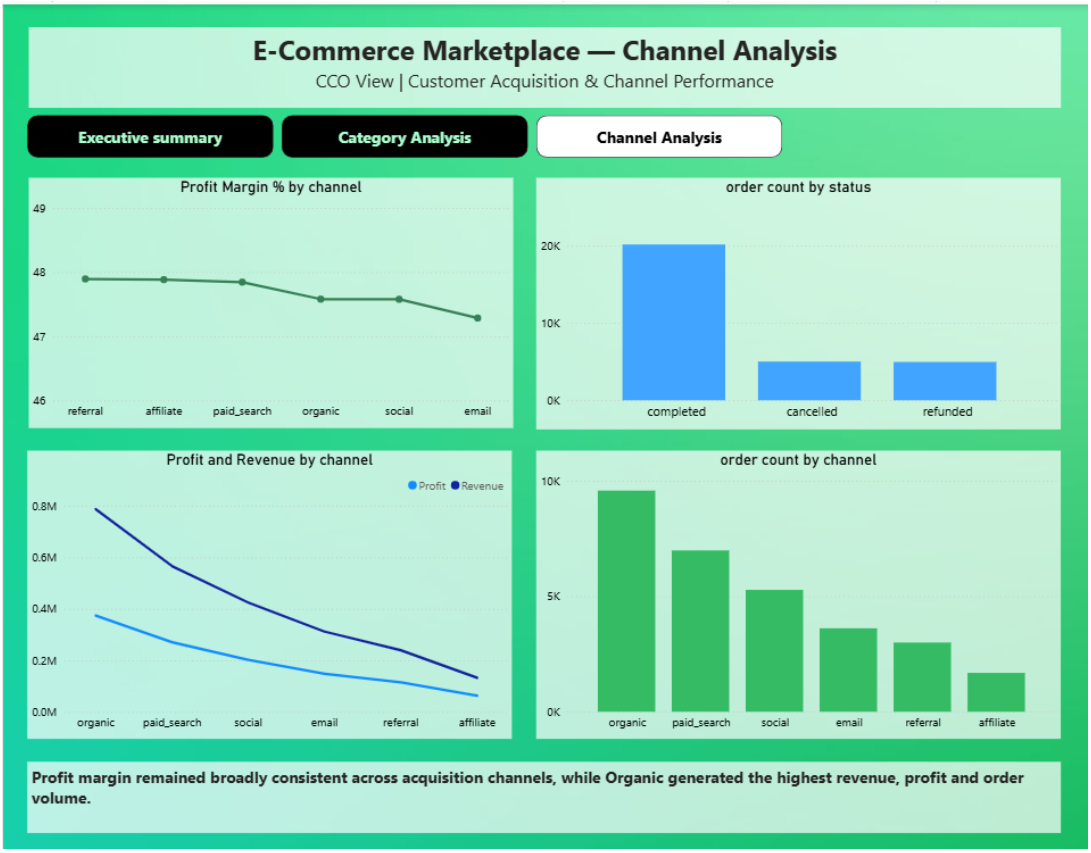
The project README identifies **SKU-0095, SKU-0051, and SKU-0038** as high-scale Outdoors products for further commercial review. The product-detail page compares revenue, profit, quantity, and profit margin, helping distinguish products that sell at scale from those that produce higher contribution or margins.

How to use this view

Use the selected category and product-level comparisons to identify high-volume products with relatively lower margins. A product with a lower margin may still generate substantial total contribution because of its scale; therefore, review both absolute profit and margin rather than using either measure in isolation.

The dashboard highlights candidates for investigation; it does not, by itself, prove that price, cost, or discounting caused the margin differences.

5. Acquisition Channel Performance



Key finding

Organic generates the highest revenue, profit, and order volume. Profit margins across acquisition channels remain broadly comparable, so margin alone is not sufficient evidence to justify reallocating channel budgets.

Commercial implication

A fuller acquisition-efficiency assessment would need marketing investment and customer-value measures. Without marketing spend, the project cannot calculate customer acquisition cost (CAC) or lifetime value (LTV). Channel scale is visible, but efficiency and return on marketing investment are not established by the available data.

6. Recommendations, Limitations & Conclusion

Recommendations

• **Review Outdoors margins:** prioritize high-scale, lower-margin products and inspect their pricing and unit costs.

• **Study high-margin products:** investigate products such as SKU-0022 for potential pricing or cost-management learnings.

• **Avoid margin-only channel decisions:** combine revenue, profit, order volume, marketing spend, and customer-value measures.

• **Extend the analysis:** add spend, CAC, and LTV measures when suitable marketing and customer data are available.

Limitations

• Marketing spend is not available, so CAC and LTV cannot be calculated.

• The Events table does not contain product or order IDs, so events cannot be directly linked to products.

• Profit represents product-level contribution based on product cost only.

• Shipping, fulfilment, and marketing costs are excluded; calculated profit should not be interpreted as net profit.

• Additional checks by country, channel, and month did not identify a clear driver of the Outdoors margin gap; the analysis identifies where performance differs, but not always why.

Conclusion

This project brings together commercial KPIs and product, category, time, and channel views in an interactive Power BI report. Its main contribution is not a claim of causality, but a structured way to identify performance differences and prioritize further investigation. The strongest follow-up is a focused Outdoors margin review, alongside richer acquisition-efficiency data for channel decisions.

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